

Pelvis and Trunk Energy Flow in Collegiate Softball Pitchers With and Without Upper Extremity Pain

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Background: The softball pitch is a full-body motion, where efficient proximal to distal energy flow through the kinetic chain is said to reduce stress at the upper extremity. Although altered trunk kinematic parameters are associated with upper extremity pain in softball pitchers, further research is needed to determine whether differences exist in proximal energy flow between softball pitchers with and without pain.

Hypothesis/Purpose: To examine pelvis and trunk energy flow during the acceleration phase of the pitch in collegiate softball pitchers with and without upper extremity pain. It was hypothesized that those with upper extremity pain would have less energy flowing into the proximal ends of the pelvis and trunk as well as less energy flowing out of the distal ends of the pelvis and trunk during the acceleration phase when compared with pitchers who did not have upper extremity pain.

Study Design: Descriptive laboratory study.

Methods: A total of 54 female National Collegiate Athletic Association Division I softball pitchers (age, 20.2 ± 2.0 years; height, 173.5 ± 6.9 cm; weight, 78.5 ± 11.5 kg) were assigned to pain ($n = 17$) and pain-free ($n = 38$) groups. Participants pitched 3 maximal effort rise-balls for a strike, and the average of the 3 trials was used for analysis. Kinematic data were collected at 100 Hz using an electromagnetic tracking system. A segment power analysis was performed to quantify energy flow for the pelvis and trunk. The Mann-Whitney U test was used to compare pelvis and trunk energy flow during the acceleration phase of the pitch as well as pitch velocity between collegiate softball pitchers with and without upper extremity pain.

Results: No significant differences were found between pelvis and trunk energy flow during the acceleration phase or pitch velocity between collegiate softball pitchers with and without upper extremity pain (all P values $>.057$).

Conclusion: Previous research determined that kinematic parameters differ between collegiate pitchers with and without upper extremity pain. However, the current study found no difference in pelvis and trunk energy flow or pitch velocity. Although altered kinematics in collegiate pitchers with upper extremity pain may serve as compensation patterns to maintain pitch velocity and proximal energy flow, future research is needed to confirm this postulation.

Clinical Relevance: The lack of energy flow differences between upper extremity pain groups suggests that pitchers may adapt their biomechanics to maintain trunk and pelvis energy flow patterns. Coaches, athletes, and clinicians should know that movement adaptations can allow for maintained performance levels but may result in the presence of pain that may manifest in other deleterious and injury-susceptible biomechanics.

Keywords: trunk; pelvis; energy flow; softball; pitchers

Successful performance of the windmill softball pitch depends on the efficient transfer of energy through the kinetic chain. Energy generated in the proximal lower extremity, pelvis, and trunk is transferred to the distal upper extremity before ball release. Decreases in proximal energy generation may result in upper extremity

compensation to maintain pitching performance.^{16,17} It is postulated that over time, these compensations may place undue stress on the upper extremity and increase the risk of upper extremity pain and injury.¹⁷ A prospective study reported that 38% of pitchers sustained an injury across a season and 72% of those injuries occurred at the upper extremity.³⁰ Pitchers are also 2.6 times more likely to sustain an injury compared with positional players.²⁹

Research has emphasized the aforementioned postulation between kinetic chain deficiencies and mechanical alterations and/or injuries in baseball and softball athletes.^{7,8,13,16} Chaudhari et al⁷ found an association between

poor lumbopelvic control and increased days missed due to injury in baseball pitchers. Friesen et al¹³ found that single-leg squat mechanics, a common assessment of lumbopelvic control, were associated with trunk mechanics during the softball pitch. Both studies highlighted the importance of the pelvis and trunk in pitching. Two studies compared mechanics between collegiate softball pitchers with and without upper extremity pain. Pitchers with upper extremity pain showed differences in trunk rotation at foot contact²³ and trunk lateral flexion at ball release.²⁶ These results led these authors to hypothesize that altered trunk kinematics may disrupt energy flow throughout the kinetic chain. However, whether pitchers differ in energy flow through segments of the kinetic chain remains unclear.

Energy flow rhetoric is commonly used in baseball and softball literature²³⁻²⁶; however, few studies have examined energy flow through the kinetic chain, especially through the proximal pelvis and trunk. Howenstein and colleagues¹⁴ indicated that energy flow into the humerus from the trunk was related to baseball pitch velocity. Aguinaldo and Escamilla³ showed that trunk segment power was a predictor of elbow valgus torque and ball velocity during the baseball pitch. In a follow-up study, Aguinaldo and Escamilla² concluded that approximately 86% of energy transfer through the elbow during the baseball pitch resulted from actions of the trunk. The only softball study to investigate energy flow during the windmill pitch showed that greater drive-hip external rotation strength was associated with increased energy flow out of the distal trunk.²⁸ Although energy flow has yet to be compared between pitchers with and without upper extremity pain, Martin et al¹⁸ determined that tennis players with an upper extremity injury had decreased ball velocity and increased energy flowing out of the trunk and into the upper extremity. The aforementioned studies have laid the foundation for investigating energy flow between softball pitchers with and without upper extremity pain.

Understanding the relationships between energy flow, trunk kinematics, and pain is vital to enhancing injury prevention programs by providing a deeper understanding of energy flow through the kinetic chain during the softball pitch. Therefore, the purpose of the current study was to compare pelvis and trunk energy flow during the acceleration phase (Figure 1) between collegiate softball pitchers with and without upper extremity pain during activity. It was hypothesized those with upper extremity pain would show reduced energy flow into the proximal pelvis and

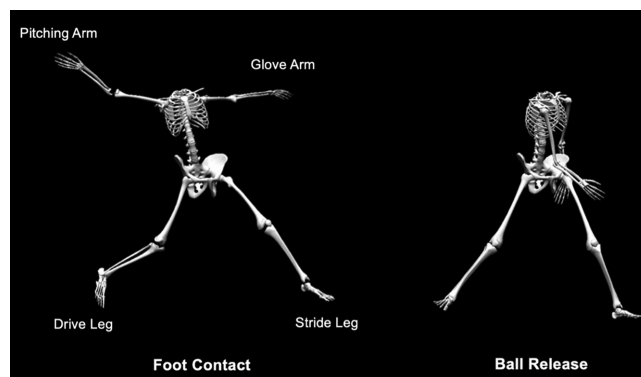


Figure 1. Acceleration phase of the pitch from foot contact to ball release.

trunk and reduced energy flow out of the distal pelvis and trunk during the acceleration phase compared with pitchers who did not have upper extremity pain.

METHODS

A total of 54 female National Collegiate Athletic Association Division I softball pitchers (age, 20.2 ± 2.0 years; height, 173.5 ± 6.9 cm; weight, 78.5 ± 11.5 kg) participated in this study. All testing procedures were approved by the Institutional Review Board of Auburn University. Inclusion criteria required participants to be actively competing on the team roster as a pitcher; additionally, they had to be surgery-free for the past 6 months. Participants arrived at the laboratory in appropriate athletic attire, all testing procedures were thoroughly explained, and informed consent was obtained.

Participants were asked, “Do you currently experience any pain/discomfort in your upper extremity, specifically your throwing side during activity?” Based on the yes/no response, participants were placed into 2 groups: those who were currently experiencing upper extremity pain and those who were not experiencing pain. Those who answered yes were deemed the pain group ($n = 17$; age, 20.6 ± 3.0 years; height, 174.9 ± 6.6 cm; weight, 84.4 ± 13.0 kg). Those who answered no were deemed the pain-free group ($n = 38$; age: 20.0 ± 1.3 years; height: 172.8 ± 7.0 cm; weight: 75.7 ± 9.8 kg).

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Kinematic and kinetic data were collected at 100 Hz using an electromagnetic tracking system (ETS) (trakSTAR; Ascension Technologies) synchronized with The MotionMonitor software (Innovative Sports Training).^{5,10,23,26,27} Previous research concluded that the ETS is reliable in measuring pelvis and trunk kinematics.²⁰ Additionally, the current study's ETS was used to quantify pelvis and trunk kinematics in several published softball pitching studies.^{9,23,26,33} We attached 14 electromagnetic sensors to the participants using previously established standards.⁴ Position and orientation of body segments were consistent with International Society of Biomechanics recommendations.^{35,36} After proper sensor attachment and system calibration, participants were provided unlimited time to perform their preferred warm-up for maximal effort pitching. Self-selected warm-ups ensured that pitchers were able to closely replicate their precompetition routine.^{26,28} Testing required participants to throw 3 maximal effort rise-ball pitches for strikes to a catcher located at regulation distance of 13.11 m (43 ft). Trials were saved for only pitches thrown within the strike zone. A calibrated radar gun (Stalker Pro II; Stalker Radar) recorded pitch velocity to the nearest mile per hour. The average of the 3 rise-ball trials was used for analysis.^{1,11,19,23} The pitching motion was examined during the acceleration phase (foot contact to ball release) (Figure 1). Only the acceleration phase was analyzed because this is when the pitching arm experiences the largest amount of stress and it is the most dynamic portion of the pitch.^{6,28,34} The pitching arm was defined as the arm used to pitch the ball, and the contralateral arm was defined as the glove arm. The drive leg was defined as the leg ipsilateral to the pitching arm, and the stride leg was defined as the leg contralateral to the pitching arm (Figure 1).

A segment power analysis was performed to quantify energy flow for the pelvis and trunk.¹⁴ Joint force power (JFP) was calculated as the dot product of the joint reaction force and joint linear velocity (Equation 1). Segment torque power (STP) was calculated as the dot product of the joint torque and segment angular velocity (Equation 2). Segment power (SP) was isolated at proximal (Equation 3) and distal (Equation 4) ends of each segment using a ground-up approach to the kinetic chain. Proximal and distal ends of each segment were defined as closer to the ground and farther up the kinetic chain, respectively. The pelvis had 2 proximal ends (stride and drive leg sides) and 1 proximal end (L5/S1 junction). The trunk had 1 proximal end (L5/S1 junction) and 2 distal ends (pitching and glove arm sides). SPs at proximal and distal ends were integrated over the time that they were negative in order to calculate total energy outflow (work done by the segment). SPs at proximal and distal ends were integrated over the time that they were positive in order to calculate total energy inflow (work done on the segment). Local minimum and maximum endpoint SPs determined the peak rates of endpoint SP outflow and inflow, respectively.

$$\text{JFP} = (\text{joint reaction force}) \times (\text{linear joint velocity}). \quad (\text{Equation 1})$$

$$\text{STP} = (\text{joint torque}) \times (\text{segment angular velocity}). \quad (\text{Equation 2})$$

$$\text{SP}_P = (\text{JFP}_{\text{proximal}} + \text{STP}_{\text{proximal}}). \quad (\text{Equation 3})$$

$$\text{SP}_D = (\text{JFP}_{\text{distal}} + \text{STP}_{\text{distal}}). \quad (\text{Equation 4})$$

Statistical analysis was performed using SPSS software (Version 28; SPSS Inc.). The alpha level was set a priori to $P < .05$. The Mann-Whitney U test was used to compare pelvis and trunk energy flow during the acceleration phase of the pitch as well as pitch velocity between collegiate softball pitchers with and without upper extremity pain. A nonparametric test was used because of a large discrepancy in sample size between pain ($n = 17$) and pain-free ($n = 38$) groups. Further, the sample was not normally distributed based on skewness scores exceeding ± 1.96 SE of skewness for all dependent variables except for total distal thorax outflow on the pitching arm side and pitch velocity. The sample was also not normally distributed based on the kurtosis scores exceeding ± 1.96 SE of kurtosis for all dependent variables except for peak rate of distal pelvis outflow, total distal trunk outflow on the pitching arm side, total proximal trunk inflow, peak rate of proximal trunk inflow, and pitch velocity. Using the Kolmogorov-Smirnov test, we determined that the dependent variables were not normally distributed (all P values $< .027$) except for total distal pelvis outflow, total distal thorax outflow on the pitching arm side, total proximal trunk inflow, and pitch velocity. Nonparametric effect sizes were estimated using the common language effect size (f), which is obtained by dividing the Mann-Whitney U statistic by the product of the group sizes (ie, $U/[n1 \times n2]$) and represents the proportion of pairs that support a certain relational direction between the groups. For example, an f value of 0.5 indicates no favoring of direction in pair comparisons (50%/50%). An f value of 0.3 indicates that 30% of pair comparisons favored group 1 (in this case, a positive pain response) having a larger value on the dependent variable.²¹

RESULTS

Descriptors of upper extremity pain are reported in Table 1. Median and interquartile range values for pelvis and trunk energy inflow and outflow are reported in Table 2. Net pelvis and trunk SP graphs during the acceleration phase of the softball pitch are presented in Figures 2 through 5. The Mann-Whitney U test indicated there were no significant differences between pelvis and trunk energy flow during the acceleration phase of the pitch as

TABLE 1
Descriptors of Upper Extremity Pain^a

Pain Descriptor	No. (%) of Pitchers
Location	
Shoulder	11 (64.7)
Elbow	10 (58.8)
Forearm	1 (5.9)
Hand	1 (5.9)
When	
During throwing	10 (58.8)
After throwing	12 (70.6)
Onset	
Gradual	11 (64.7)
Sudden	4 (23.5)
Not specified	2 (11.8)

^aThe sums of percentages are >100 for "Location" and "When" because several participants had pain in multiple locations and experienced pain during and after throwing.

well as pitch velocity between collegiate softball pitchers with and without upper extremity pain (all *P* values >.057; all *f* values between 0.34 and 0.47).

DISCUSSION

The purpose of this study was to compare acceleration phase pelvis and trunk energy flow between collegiate softball pitchers with and without upper extremity pain. Of particular interest were the total amounts of inflow and outflow (Total IF and OF in Table 2) and peak rates of inflow and outflow (Peak IF and OF in Table 2) to and from the pelvis and trunk during the acceleration phase of the windmill pitching motion. It was hypothesized that those with upper extremity pain would show reduced energy flow into the proximal ends of the pelvis and trunk

and reduced energy flow out of the distal ends of the pelvis and trunk. This hypothesis was rejected because we found no significant differences in total pelvis and trunk energy flow or peak rate of energy flow (power) between pitchers with and without upper extremity pain.

Softball pitchers currently experiencing upper extremity pain have shown increased trunk rotation toward the pitching arm at foot contact and decreased trunk lateral flexion toward the pitching arm at ball release.^{23,26} Therefore, we anticipated that pelvis and trunk energy flow would also differ between pitchers with and without pain. Although we observed no significant differences in energy flow or pitch velocity between pain groups, null findings provided evidence that altered kinematics in the upper extremity pain group may serve as a compensation mechanism to maintain energy flow between segments and pitching performance. Softball pitchers have a high incidence of upper extremity pain and are accustomed to pitching through pain in competition settings.³² Soldatis et al³¹ determined that 47% of softball athletes play through in-season upper extremity pain. Although previous research reported kinematic differences between collegiate pitchers with and without upper extremity pain,^{23,24,26} we found no differences in pelvis and trunk energy flow in the current study's population of collegiate pitchers. Together, these studies suggest that altered kinematics in collegiate pitchers with upper extremity pain may be compensation patterns used to maintain pitch velocity and proximal energy flow. Further, research has found that kinematic alterations can negatively influence kinetics, so one might assume this relationship could explain the reported pain.^{12,15,26} However, a future study should examine both kinematics and proximal energy flow across collegiate pitchers with and without upper extremity pain to test this postulation.

Although pitchers with and without upper extremity pain in the current study showed similar total amounts

TABLE 2
Median and Interquartile Range Values for Pitchers With and Without Upper Extremity Pain^a

Variable	Pain (n = 17)	Pain-Free (n = 38)	<i>P</i> Value	<i>f</i>
Pelvis				
Total proximal IF _S	142 (66-209)	163 (113-183)	.39	0.43
Total proximal IF _D	42 (24-57)	45 (34-67)	.38	0.43
Total distal OF	165 (114-241)	190 (143-239)	.27	0.41
Peak proximal IF _S	3404 (1885-4985)	3580 (2437-4885)	.68	0.46
Peak proximal IF _D	1156 (923-1433)	1430 (979-1911)	.08	0.35
Peak distal OF	3905 (2630-5375)	4336 (3501-5875)	.21	0.39
Trunk				
Total proximal IF	133 (61-207)	164 (98-275)	.18	0.39
Total distal OF _{PA}	112 (57-144)	107 (55-146)	.70	0.47
Total distal OF _{GA}	2 (0-9)	6 (1-19)	.06	0.34
Peak proximal IF	3583 (2642-5755)	4893 (3328-6737)	.16	0.38
Peak distal OF _{PA}	2066 (1451-2624)	1980 (1426-2848)	.81	0.48
Peak distal OF _{GA}	86 (31-324)	235 (73-848)	.12	0.37
Pitch speed, mph	53 (49-56)	54 (53-56)	.09	0.36

^aData are presented as median (interquartile range). Total IFs and OFs are presented in joules; peak IFs and OFs are presented in watts. D, drive leg side; GA, glove-arm side; IF, inflow; OF, outflow; PA, pitching arm side; S, stride leg side.

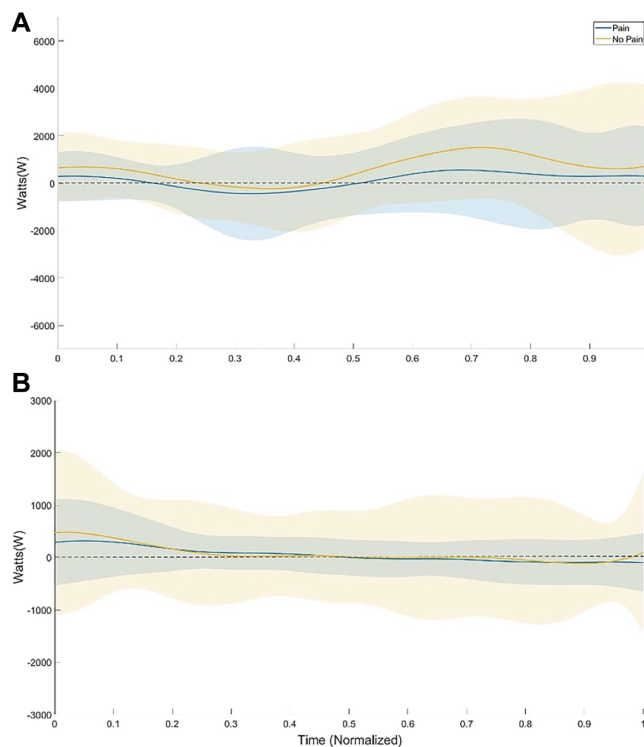


Figure 2. Proximal pelvis segment power on (A) stride and (B) drive hip sides during the acceleration phase of the windmill softball pitch between pitchers with and without upper extremity pain. Watts >0 indicate energy flow into the segment and <0 indicate energy flow out of the segment. Time 0 = foot contact; time 1 = ball release.

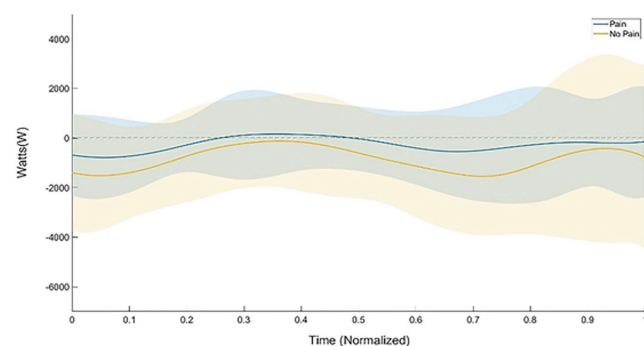


Figure 3. Distal pelvis segment power during the acceleration phase of the windmill softball pitch between pitchers with and without upper extremity pain. Watts >0 indicate energy flow into the segment and <0 indicate energy flow out of the segment. Time 0 = foot contact; time 1 = ball release.

and peak rates of energy flow through the pelvis and trunk, it is still unclear what happens to the energy after it leaves the trunk and enters the pitching arm. Although increased energy flow out of the trunk to the pitching arm could be considered a positive factor for pitching

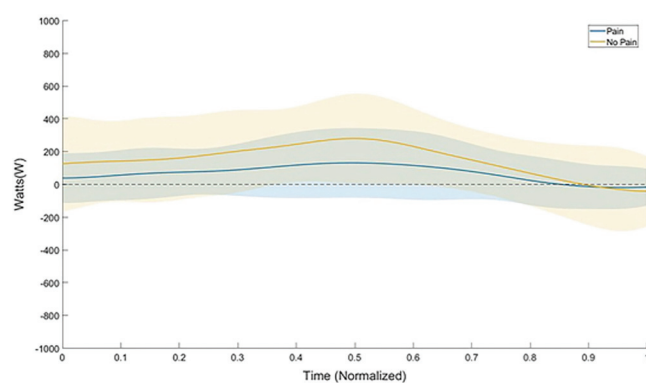


Figure 4. Proximal trunk segment power during the acceleration phase of the windmill softball pitch between pitchers with and without upper extremity pain. Watts >0 indicate energy flow into the segment and <0 indicate energy flow out of the segment. Time 0 = foot contact; time 1 = ball release.

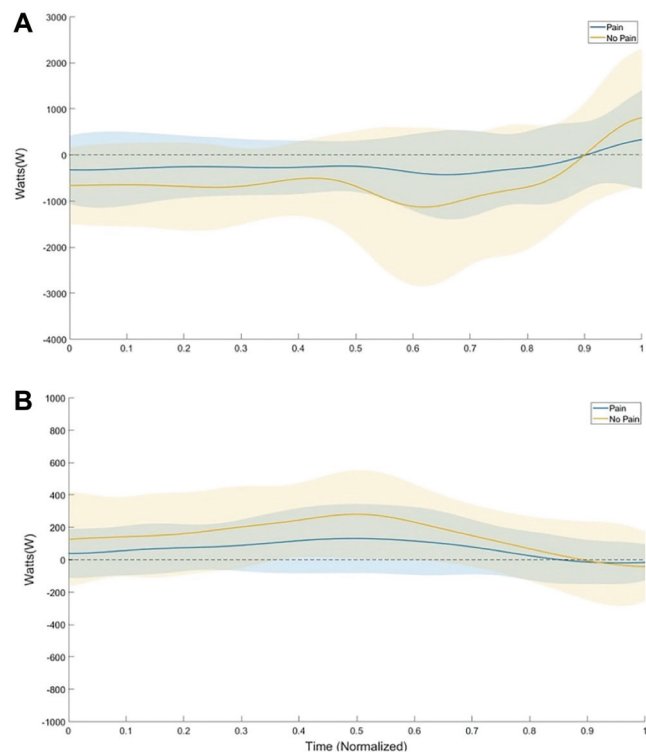


Figure 5. Distal trunk segment power on the (A) pitching and (B) glove arm sides during the acceleration phase of the windmill softball pitch between pitchers with and without upper extremity pain. Watts >0 indicate energy flow into the segment and <0 indicate energy flow out of the segment. Time 0 = foot contact; time 1 = ball release.

performance, it can also be a negative factor if that increased energy outflow is not adequately transferred through the upper extremity. The importance of energy flow into the upper extremity is supported by the findings

from Martin et al,¹⁸ who reported that tennis players without an upper extremity overuse injury had higher rates of energy leaving the trunk and entering the upper extremity compared with players without an injury. From a performance perspective,¹⁸ Howenstein and colleagues¹⁴ showed that energy flow into the pitching arm had the greatest association with baseball pitch velocity.

Although we found no group differences, this study adds valuable information by being one of the first to provide energy flow data in collegiate softball pitchers. A descriptively larger amount of total and peak proximal energy inflow to the pelvis was from the stride leg (total: pain group, 142 J; pain-free group, 163 J; peak: pain, 3404 W; pain-free, 3580 W) as opposed to the drive leg (total: pain, 42 J; pain-free, 45 J; peak: pain, 1156 W; pain-free, 1430 W). Energy flow was assessed during the acceleration phase (foot contact to ball release) (Figure 1), which excludes the push-off phase of movement. This may explain a descriptively smaller amount of energy inflow to the pelvis from the drive leg. Optimal push-off mechanics have been positively associated with performance, and decreased drive leg strength has been observed in pitchers with upper extremity pain compared with those who have no pain.²⁵ Thus, future research should also consider analyzing energy flow during the push-off phase.²²

Limitations

Although this study contributes to the literature on softball pitching energy flow, limitations do exist. The current study analyzed energy flow of the pelvis and trunk during the acceleration phase of the softball pitch. Future research should compare upper and lower extremity energy flow parameters during all phases of the pitch between softball pitchers with and without upper extremity pain. Additionally, the current study calculated only the net proximal and distal SPs. A closer examination of energy generation, absorption, and transfer can be made by isolating and partitioning the respective STPs and JFPs. It is theoretically possible that overall energy flow patterns are similar yet the relative proportions of energy generation and/or absorption differ between pitchers with and without upper extremity pain. Future work should investigate generation and absorption patterns to elucidate this possible difference. The presence of pain was indicated as a binary response. A sliding scale of pain severity may offer greater insight into the influence of pain and how it might cause biomechanical differences while pitching. Last, energy flow parameters were measured during the presence of upper extremity pain. It is unknown whether the observed values would be similar to those measured before the onset of upper extremity pain.

CONCLUSION

We found no significant differences in pelvis and trunk energy flow during the acceleration phase or pitch velocity between pitchers with and without upper extremity pain. Although no group differences were found, this study adds valuable information by being one of the first, to

our knowledge, to provide energy flow data in collegiate softball pitchers. The results suggest that altered trunk kinematics previously reported in collegiate pitchers with upper extremity pain may serve as compensation patterns to maintain pitching performance and energy flow between segments.^{23,26}

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